

ADC SOLUTIONS

Project Deschutes Coolant Distribution Unit (CDU)

Procurement Bill of Materials / Bill of Design Package

Version 0.1 - With Four Explanatory Diagrams

Document No.	ADC-CDU-DES-BOM-001
Date	May 19, 2026
Prepared for	ADC Solutions
Scope	Deschutes-aligned CDU procurement BOM, including internal CDU subsystems and CDU-side interface allowances.
Architecture Control	No pump resize. No main Deschutes CDU architecture change. Pump size, HX topology, primary valve topology, and 2 MW / 500 GPM design basis are preserved.
Classification	CONFIDENTIAL - ADC Solutions Internal & Authorized Vendors Only
Engineering lock statement This BOM intentionally preserves the Project Deschutes main CDU architecture: two sealless pumps in N+1 service, three two-pass heat exchangers in parallel, one primary-side control valve, 2 VFDs, 500 GPM IT/secondary flow, 500 GPM facility/primary flow, and 2 MW thermal capacity.	

Prepared from uploaded reference documents: OCP Project Deschutes v1.0, ADC Modular TCS BoD/RFQ v1, Deschutes Redfish/AI/TCS Reference v1.2, Open Data Center v0.5.0, and Diablo 400 v0.5.2.

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1. Design Basis and Architecture Locks

Purpose. This package converts the supplied Project Deschutes and ADC TCS guidance into a CDU procurement BOM. It is a procurement / RFQ-level BOM, not a released manufacturing BOM. Exact fastener counts, spool lengths, cabinet harness lengths, and bracket part numbers shall be reconciled to final supplier CAD and supplier submittals.

Thermal capacity	2,000 kW (2 MW) per CDU
Secondary / IT loop flow	500 GPM design point
Facility / primary loop flow	500 GPM at 0.25 GPM/kW
Secondary available pressure	80-90 psi available at CDU secondary outlet
Facility pressure drop basis	15-30 psi
Operating pressure	0-130 psig; pressure relief at 130 psi per Deschutes product requirements
Operating liquid temperature	18-55 deg C
Approach temperature target	3 deg C across the liquid-to-liquid heat exchanger at design load
Coolants	DI water and PG25; wetted materials shall be compatible with approved inhibited DI-water coolants and PG25
Wetted material policy	Stainless steel, copper, EPDM; nickel-plated brass not allowed on the secondary side
CDU physical basis	Approx. 65.25 in W x 91.44 in H x 42.29 in D; wet max 6,900 lb; dry max 5,310 lb. Final rigging data to supplier submittal.
Power basis	380-416 Vac, 3-phase, 50/60 Hz; dual feed PDU; total CDU power approx. 74 kW

Source basis: OCP Project Deschutes v1.0 Sections 3.1, 3.2, 5.1-5.3, 6.1-6.7, and 7.1-7.4; ADC Modular TCS BoD/RFQ Section 2.1-2.2.

Non-negotiable procurement controls

1) Do not substitute or resize the pump without a formal deviation. 2) Do not change from the Deschutes three-HX parallel architecture. 3) Do not move TCS rack-loop safety authority out of the PLC. 4) Do not use nickel-plated brass on the secondary side. 5) Treat all quantities marked "allowance" as CAD-release items rather than architecture changes.

2. Scope Boundaries

Boundary	Included in this CDU BOM	Excluded / handled by separate package
Inside CDU cabinet	Pumps, HXs, valve, filters, air separator, expansion, internal piping, sensors, PLC/HMI, VFDs, PDU, service hardware.	Final production-level spool drawings, harness drawings, and work instructions are supplier CAD deliverables.
CDU-to-TCS connection	CDU-side connection allowance, secondary-side side-kit interface, instrumentation handoff, leak-zone inputs, TCS data gateway allowance.	Full 6 in TCS header, DN50 branch lines, rack QDs, rack manifolds, cold plates, and row drip trays remain in the modular TCS BoD/RFQ package.
Primary / facility side	CDU primary control valve, PCW manifold, primary filter, primary sensors, facility-side connection kit allowance.	Chiller plant, cooling towers/dry coolers, facility pumps, BMS controls, and upstream PCW distribution.
Power / sidecar	CDU PDU, VFD power, GFCI, local disconnects, control power, grounding.	Diablo 400 sidecar / HVDC power rack BOM and IT-rack power distribution.
Controls	PLC native Modbus/TCP, HMI, sensor I/O, optional Redfish gateway, telemetry mapping.	Enterprise DCIM, workload scheduler, external AI platform, and production cybersecurity stack.

Source basis: Scope follows ADC Modular TCS BoD/RFQ Sections 1.1-1.3 and Deschutes Redfish/AI/TCS Reference Section 4.5.

3. BOM Coding Convention

Prefix	Subsystem	Notes
CDU	Cabinet, frame, manifolds, internal piping	Base CDU mechanical structure and common hardware.
PMP	Pump train	SPECK TOEMA065200 pumps and service hardware; pump size locked.
HX	Heat exchanger bank	Alfa Laval CB210-276AH x3 in parallel; two-pass.
PV	Primary-side valve and primary piping	Belimo B6500 primary control valve and supporting valves.
FLT	Filtration / microfiltration	Main filters and 0.2 micron side-stream microfilter.
AIR	Air removal / deaeration	Spirovent air separator and vents.
EXP	Expansion / reservoir	Flow-through expansion tank and service kit.
SEN	Instrumentation	Pressure, temperature, flow, coolant quality.
LEAK	Leak detection	Three-zone leak rope and controller to PLC.
PLC	Controls	PLC, HMI, I/O, firmware, Modbus/TCP.
RF	Redfish / AI bridge	Optional gateway if Redfish is not directly implemented in the controller.

Prefix	Subsystem	Notes
PWR	Electrical power distribution	PDU, disconnects, fuses, wiring, grounding.
VFD	Variable frequency drives	ABB ACH580 per pump.
SK	CDU-side TCS interface allowance	Interface only; row TCS header and branches stay in TCS BOM.
VAL	Validation / QA / documentation	Pressure, leak, thermal-hydraulic, manufacturing and FRU deliverables.

4. Detailed CDU BOM

4.1 Cabinet, Frame, Manifolds, and Internal Piping

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
CDU-01	CDU base frame and enclosure	1 set	Deschutes CDU form factor; front/rear service access; max packaged height 100 in.	Include lifting/rolling provisions, removable service panels, access clearances, labels, and safety guards. Final dimensions by supplier CAD.
CDU-02	Secondary rack water supply/return manifolds	1 set	Stainless/copper/EPDM wetted material policy; 0-130 psig operating; PG25 and DI-water compatible.	Manifold names: rack water supply and rack water return. Include pressure/temperature taps.
CDU-03	Primary PCW supply/return manifolds	1 set	Facility water inlet/outlet manifold set compatible with DI water and approved inhibitors.	Include Belimo primary valve integration and facility-side sensor taps.
CDU-04	Internal stainless piping and fittings lot	1 lot	304/316/316L stainless; grooved, tri-clamp, welded, or brazed joints per supplier design; pressure-rated for Deschutes system.	Use polished/passivated stainless practices; verify weld heat tint acceptance and cleanliness.
CDU-05	Victaulic / tri-clamp coupling and gasket kit	1 lot	EPDM gaskets; stainless or compatible metallic clamps; optimized mix of Victaulic and tri-clamp fittings.	Use serviceable joints where FRU removal is required; avoid unnecessary pressure drop.
CDU-06	Internal drip / leak capture provisions	1 set	Non-corroding containment under leak-prone components; tie into leak detection or inspection path.	Coordinate with three-zone leak detection and cabinet service access.
CDU-07	Barcode, nameplate, and safety label set	1 lot	Component ID, flow direction, voltage, relief pressure, coolant, QR/barcode labels, service labels.	Follow Deschutes barcode / component traceability intent and ADC procurement practice.

4.2 Pump Train - Pump Size Locked

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
PMP-01	Secondary-loop pump	2	SPECK TOEMA065200; 500 GPM at 112 psi; 37 kW or 50 HP; sealless; N+1 operation.	LOCKED ITEM. Do not resize, remove, or substitute without formal engineering deviation.
PMP-02	Pump mounting and slide/service hardware	2 sets	Low-friction pads or service slides to allow pump movement and removal from front/side as Deschutes requires.	Coordinate with pillar interference and service access drawings.
PMP-03	Pump inlet/outlet isolation valves	4	Full-port, coolant-compatible valves at each pump inlet and outlet.	Required to isolate redundant pump during maintenance.
PMP-04	Pump check valves / backflow prevention	2	Check valves where required to prevent reverse flow through inactive pump path.	Final valve type and Cv per hydraulic model.
PMP-05	Pump drain ports and bleed points	4 min	Drain/vent hardware at pump inlet/outlet low/high points; stainless/EPDM wetted materials.	Supports FRU replacement and commissioning.

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
PMP-06	Pump power disconnect / connectorized power kit	2 sets	Finger-safe, serviceable connectorized power or breakout box; interlocked where necessary.	Coordinate with VFD, power panel, and service procedures.
PMP-07	Pump telemetry integration	2 sets	Pump speed RPM/percent, pump power kW, fault status, over-temperature alarms.	Expose to PLC/HMI and Modbus/Redfish mapping as applicable.

Source basis: Project Deschutes Sections 5.4, 6.1, 7.2, 7.3, 13.2, 20.13, and 23.1.3.

4.3 Heat Exchanger Bank and Primary-Side Control

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
HX-01	Liquid-to-liquid heat exchanger	3	Alfa Laval CB210-276AH; two-pass; three heat exchangers in parallel.	Do not change HX count/topology without deviation. Parallel bank is used for lower pressure drop and thermal performance.
HX-02	HX parallel manifold kit	1 set	Primary and secondary distribution manifolds for equal flow across three HXs.	Supplier to provide balance/flow distribution basis and dP model.
HX-03	HX service isolation, drains, and vents	1 set	Isolation and drain/vent provisions for HX bank service and commissioning.	Exact quantities per CAD; avoid trapped air and dead legs.
PV-01	Primary-side flow control valve	1	Belimo B6500 with actuator; primary/facility side of CDU.	Controls PCW flow to maintain approach-temperature target.
PV-02	Primary-loop isolation valves	2 min	Facility supply/return isolation valves; rated for operating pressure, temperature, and coolant chemistry.	Coordinate with facility water pipe kit and AHJ requirements.
PV-03	Primary-loop check/backflow valves	Allowance	Check valves where appropriate to prevent reverse flow.	Valve locations per final hydraulic schematic.
PV-04	Primary valve command/feedback wiring	1 set	ValvePositionPercent / ValvePositionControlPercent feedback to PLC/HMI.	Required for controls, Redfish mapping, and AI supervisory setpoint model.

Source basis: Project Deschutes Sections 5.3, 6.2, 6.3, 7.2.1 and Redfish/AI/TCS Reference Sections 4.4 and 8.3.

4.4 Filtration, Air Management, Expansion, Fill/Purge, and Coolant Quality

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
FLT-01	Primary main filter	1	Front-serviceable 25-44 micron filter assembly; DI/PG25 compatible.	Primary/facility side filter. Include differential pressure ports/sensor.
FLT-02	Secondary main filter	1	Front-serviceable 25-44 micron filter assembly; DI/PG25 compatible.	Secondary/IT side filter. Include differential pressure ports/sensor.
FLT-03	Side-stream microfiltration loop	1 set	0.2 micron side-stream/bypass filtration, approximately 3 GPM, front serviceable.	Include isolation valves, service drain, dP monitoring, and replacement element allowance.
AIR-01	Air separator	1	Spirovent air separator or equivalent; compatible with operating pressure/temp and coolant.	Include high-point venting and commissioning purge support.
EXP-01	Flow-through expansion tank	1	Front-serviceable flow-through expansion tank; bladder/diaphragm materials compatible with PG25/DI water.	Size to supplier hydraulic model; service isolation and pressure monitoring required.
EXP-02	Expansion tank service kit	1 set	Isolation, drain, pressure gauge/transducer, and test fittings.	Supports impedance, leak, pressure-cycle, and system functional validation.

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
FILL-01	Fill and purge port set	1 set	Fill/purge ports for both primary and secondary loops as required by commissioning procedure.	Use stainless wetted hardware; include caps, chains, and labeling.
PRV-01	Pressure relief valve	1 min	Pressure relief at 130 psi; UL suitable component or ASME/UV stamped component as applicable.	Final quantity and loop placement by safety review and AHJ; coordinate with CDU OEM.
CQT-01	Coolant pH/conductivity telemetry	1 set	pH and electrical conductivity sensors/analyzer; coolant-compatible wetted materials.	Tie to PLC/HMI and telemetry namespace.

Source basis: Project Deschutes Sections 3.2, 5.2, 8.2, 20.5-20.10, and ADC Modular TCS BoD/RFQ Section 2.2.

4.5 Instrumentation, Leak Detection, Controls, and Redfish/AI Interface

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
SEN-01	Combined pressure and temperature sensors	4 min	Sensors at rack water supply, rack water return, PCW supply, PCW return; calibrated to datasheet accuracy.	Primary feedback for hydraulic and thermal controls.
FLOW-01	Secondary / technical loop flow meter	1	Flow meter with +/-0.5% of reading accuracy; compatible with DI water and PG25.	Front-serviceable; service may require CDU power-off per Deschutes service note.
SEN-02	Filter differential pressure sensors	2-3	dP sensors across primary filter, secondary filter, and microfilter loop if required.	Detects fouling and supports Redfish/TCS coolant-quality view.
LEAK-01	Leak detection sensing cable	3 zones	Water-sensing rope/cable split into three zones and routed inside CDU and interface areas.	Detected through PLC; coordinate with external TCS leak zones.
LEAK-02	Leak detection controller / I/O module	1	Controller with dry contact and/or Modbus/BACnet/PLC I/O outputs.	PLC shall validate functionality with leak detection system.
PLC-01	PLC and HMI	1	Industrial PLC with HMI; Modbus/TCP; IPv4 and IPv6 dual-stack; supports pump, valve, flowmeter and P/T telemetry.	Allen-Bradley CompactLogix 5380 or equivalent is acceptable for POC; supplier may propose equivalent production PLC.
PLC-02	Remote I/O and terminal blocks	1 lot	Analog inputs, discrete I/O, network I/O, safety interlocks, service terminals.	Include spare I/O capacity for TCS rack-loop points and coolant-quality telemetry.
RF-01	Redfish bridge / industrial gateway	1 optional	Industrial PC or embedded service translating Modbus/TCS data to Redfish-style REST/JSON endpoints.	Required if PLC/controller does not expose Redfish directly. PLC remains safety authority.
NET-01	Controls Ethernet network kit	1 set	Industrial Ethernet switch, patching, cybersecurity-ready network segmentation provisions.	Support Modbus/TCP, Redfish bridge, HMI, and monitoring laptop/SCADA handoff.
SW-01	Controls firmware and telemetry map	1 package	PLC logic, HMI screens, Modbus register map, Redfish/OEM sensor map, backup image.	Vendor must provide firmware update procedure and controls validation evidence.

Source basis: Project Deschutes Sections 6.5, 6.7, 7.2.1-7.2.3, 20.9-20.15, 21.1.4-21.1.6; Redfish/AI/TCS Reference Sections 2.2, 3.1-3.2, 4.4-4.5.

4.6 Electrical Power, VFDs, Protection, and Grounding

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
PWR-01	Dual-feed CDU PDU	1	Dual-feed 416 Vac-class PDU; 380-416 Vac nominal, 3-phase, 50/60 Hz, 1N minimum feed redundancy, 2N desired.	Coordinate with site power scheme and UL/CE listing strategy.

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
VFD-01	Pump VFD	2	ABB ACH580-31-065A-4 with +B056 NEMA Type 12 option; one VFD per pump; 50 HP / 37.3 kW class.	Matched to locked pump size and Deschutes VFD basis.
PWR-02	VFD fuse/disconnect kit	2 sets	Initial 80 A fast-acting Class J/gG fuse selection; fuse blocks/disconnects rated to 100 A and SCCR 100 kA.	Use 100 A max backup only if inadvertent tripping occurs, per Deschutes power summary.
PWR-03	Ground-fault protection	1 set	GFCI/GFPE 10-30 mA as applicable; 20 mA noted in Deschutes safety requirements.	Coordinate with UL/CE listing and site power system.
PWR-04	24 Vdc control power supply and distribution	1 set	Redundant or protected control power for PLC, sensors, leak controller, HMI, network hardware.	Coordinate hold-up behavior and power-loss alarms.
PWR-05	Internal harnesses, cable tray, gland plates	1 lot	VW-1 cabling; CL2P for water-sense cable where required; IP66 connectors and IP55 pump/motor/drive ingress targets.	Maintain separation between power and controls. Include EMC/grounding details.
PWR-06	Grounding and bonding kit	1 lot	Ground continuity per safety standards; bonding jumpers for doors, panels, frames, pipe sections, and electrical enclosures.	Include test points and final ground-bond test report.

Source basis: Project Deschutes Sections 7.1-7.4 and 8.1-8.2; Diablo 400 referenced only for data-center sidecar context, not as a CDU internal power architecture change.

4.7 CDU-to-TCS Interface Allowances

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
SK-01	Secondary-side side kit / CDU-to-TCS connection allowance	1 set	Hoses/fittings/isolation valves/instrumentation between CDU secondary ports and TCS header takeoff.	Interface only. Full TCS header and branch BOM remains separate; do not alter 6 in header or DN50 branch basis from ADC TCS package.
SK-02	Secondary-side interface instrumentation	1 set	Pressure/temperature ports and optional flow/Delta-P points at CDU/TCS handoff.	Supports commissioning, balancing, and rack-loop telemetry.
SK-03	TCS telemetry gateway allowance	1 optional	Modbus/TCP, BACnet/IP, MQTT, SCADA, or Redfish/OEM sensor bridge for branch flow, branch Delta-P, rack supply/return temperature, balancing valve position, leak zones, and coolant quality.	Use only if TCS sensors are not directly wired into CDU PLC or site SCADA.
SK-04	Hotswap telemetry quick disconnects	4 allowance	Parker NSP-251 with ORB6 fitting or equivalent for pressure telemetry quick disconnects.	Final count per pressure sensor/tubing arrangement.
SK-05	TCS leak-zone handoff terminals	1 set	Terminals or network points for CDU/TCS leak-zone alarms into PLC/BMS.	Coordinate leak response authority with site safety logic.

Source basis: ADC Modular TCS BoD/RFQ Sections 1.1-1.3, 2.1, 4.6, and 5.1-5.3; Deschutes Redfish/AI/TCS Reference Section 4.5.

4.8 Validation, Quality, Serviceability, and Documentation

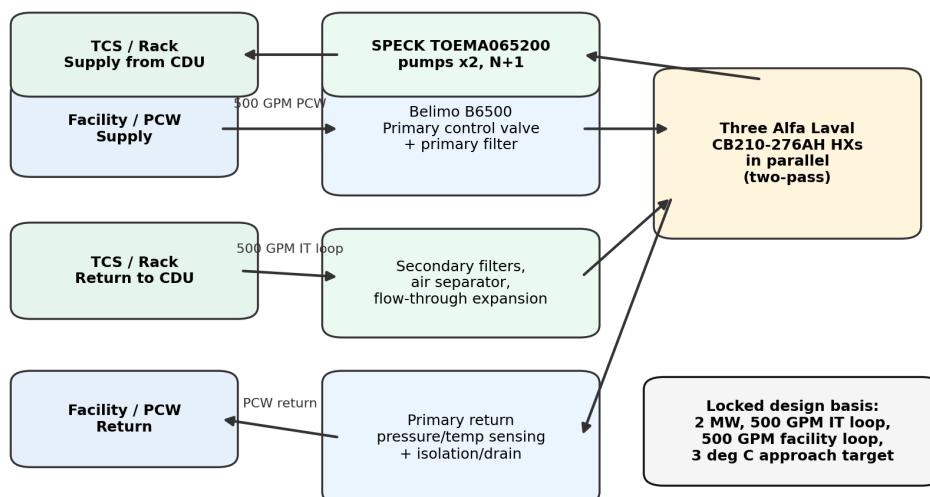
Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
VAL-01	Hydrostatic / pressure test package	1 package	Full system pressure testing, component pressure hold/cycle, and burst validation where required.	Validate with DI water and PG25 as applicable. Include pressure logging.

Tag	Item	Qty / CDU	Required spec / first source	Procurement notes / source basis
VAL-02	Helium leak / N2 pressure test package	1 package	N2 high-pressure test, helium leak test, visual inspection, alignment checks.	Coordinate with manufacturing qualification and outgoing quality control.
VAL-03	Thermal-hydraulic validation package	1 package	Hydraulic curves at 25/40/60/80/100% pump speeds; primary valve opening matrix; thermal tests up to 2 MW and beyond if available.	Include one-pump and two-pump operation; validate with DI water and PG25.
VAL-04	Pump and VFD reliability tests	1 package	Pump temperature shutoff, motor lock, non-ideal power tests, long-term fluid temperature / PG25 / VFD conditions.	Required because pump performance and energy depend on fluid type and temperature.
VAL-05	PLC/HMI/Modbus functional validation	1 package	Pump command/feedback, valve command/feedback, flowmeter feedback, pressure/temperature telemetry, firmware update, Modbus over IPv4/IPv6.	Demonstrate all functions via HMI and Modbus.
VAL-06	FRU and maintenance procedure package	1 package	Pump, filter, flowmeter, VFD, expansion tank, control panel service procedures and ergonomic review.	Document service steps, interlocks, safe isolation, and required tools.
VAL-07	Seismic / shock / vibration / COG package	1 package	COG analysis, FEA, thermal shock/low-temp exposure, non-operational and operational shock/sine, seismic considerations.	Use site-specific seismic criteria for Louisiana installation; Deschutes West-Coast Sds is conservative reference only.
DOC-01	Supplier submittal package	1 package	CAD, P&ID, electrical one-line, controls map, wiring diagrams, datasheets, material certs, cleaning/passivation records, FAI/CpK data, OQC data.	Release gate for purchase order and factory acceptance.
PKG-01	Packaging and shipping package	1 package	Packaging height/rigging controls, nitrogen charge retention, shelf-life requirements, protection of clean wetted parts.	Ship sealed and clean to minimize jobsite flush burden.

Source basis: Project Deschutes Sections 19-23 and ADC Modular TCS BoD/RFQ Section 2.5.

5. Four Diagrams for the Most Difficult Concepts

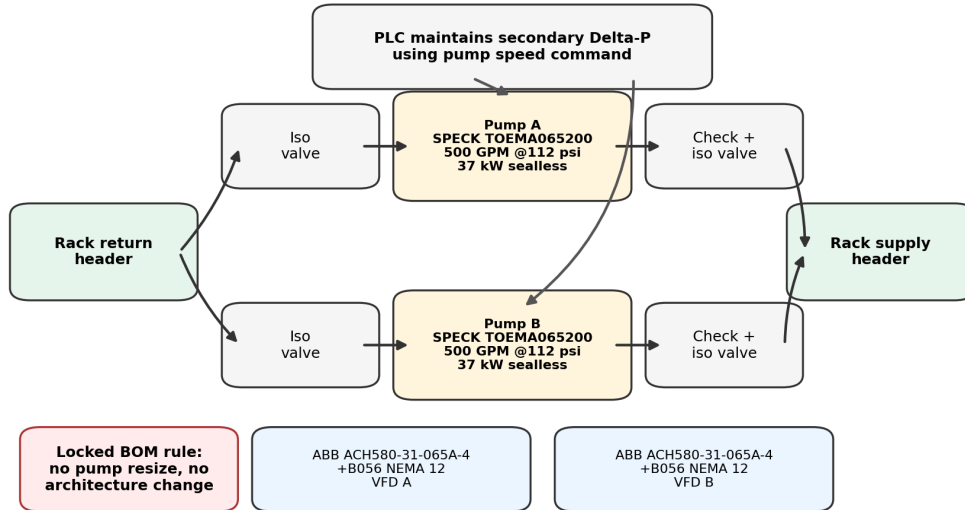
Diagram 1 - Deschutes CDU Dual-Loop Architecture (Architecture Preserved)



Primary/facility loop and secondary/IT loop are separated by the liquid-to-liquid HX bank. This BOM does not alter the Deschutes pump size, HX count, or main CDU topology.

Figure 1. Dual-loop CDU architecture: primary/facility loop and secondary/IT loop remain separated by the three-HX bank.

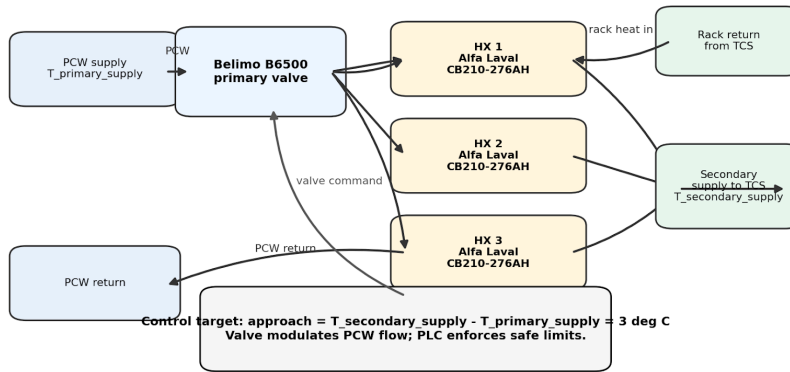
Diagram 2 - N+1 Pump Train, VFD Control, and Service Isolation



Both pumps remain the Deschutes-specified size. Service isolation, drains, and connectorized power support FRU replacement without redefining the hydraulic architecture.

Figure 2. N+1 pump train: two locked-size SPECK TOEMA065200 pumps, one VFD per pump, with isolation and service breaks.

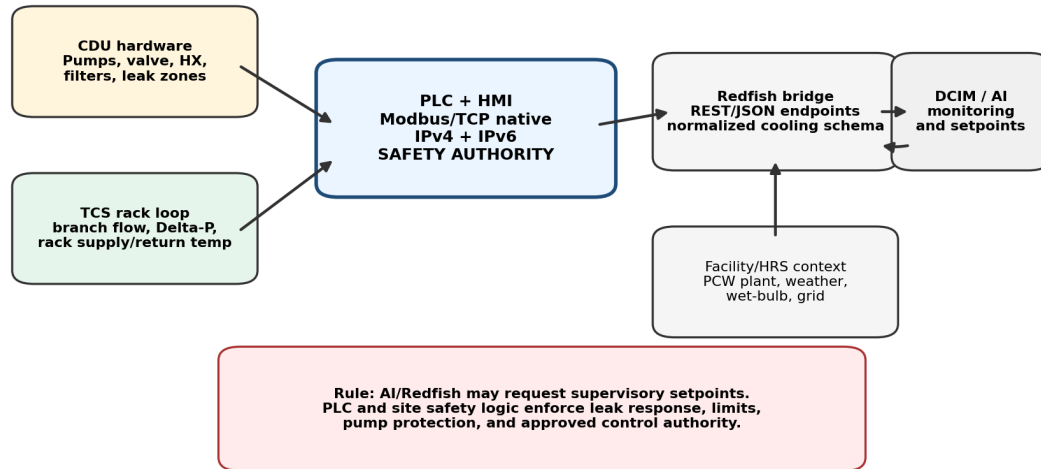
Diagram 3 - Three-HX Parallel Bank and 3 deg C Approach Control



The parallel HX bank lowers pressure drop and keeps the main Deschutes heat-transfer architecture intact. The BOM includes the valve, sensors, isolation, and service hardware required to operate this topology.

Figure 3. Three-HX parallel bank and 3 deg C approach control using the Belimo B6500 primary valve.

Diagram 4 - PLC Safety Boundary, Redfish Bridge, and TCS Rack-Loop Telemetry



The BOM includes sensors, PLC, network interface, and optional Redfish gateway to map CDU and TCS measurements without unsafe direct control.

Figure 4. PLC safety authority with optional Redfish bridge and TCS rack-loop telemetry integration.

6. Vendor Deliverables and Validation Requirements

Deliverable	Minimum content	Release gate
P&ID and hydraulic schematic	Primary loop, secondary loop, HX bank, pump train, filters, air separator, expansion, relief, vents/drains, sensor IDs, and service isolation.	Required before component procurement release.
Electrical one-line and panel drawings	Dual-feed PDU, VFD feeders, fuses/disconnects, GFCI/GFPE, control power, bonding, SCCR, ingress ratings.	Required before electrical fabrication.
Controls / telemetry register map	PLC I/O, HMI, Modbus/TCP register map, IPv4/IPv6 configuration, Redfish/OEM mapping if included.	Required before FAT controls testing.
Thermal-hydraulic model	Pump curve overlay, HX dP/approach, filter dP, facility valve authority, one-pump and two-pump cases, DI and PG25 cases.	Required before factory validation plan approval.
Material compatibility package	Wetted material list, elastomer certs, coolant compatibility, no nickel-plated brass on secondary, cleaning/passivation records.	Required before first article approval.
Factory acceptance test report	Hydraulic curve, thermal approach, pressure test, helium leak, pump/VFD functional, PLC/HMI, leak detection, flow meter accuracy, sensor calibration.	Required before shipment.
Service / FRU procedure package	Pump, filters, microfilter, flow meter, VFD, expansion tank, and control panel service procedures with tools and safe-isolation steps.	Required before site acceptance.
As-built data package	As-built CAD, marked-up P&ID, serial numbers, barcode records, calibration certs, firmware versions, spare-parts list.	Required for handover.

7. Open Items / Release Gates

ID	Open item	Owner / close condition
ARCH-01	Confirm that every supplier substitution preserves pump model/size, HX count/topology, and primary valve function.	ADC + CDU supplier; close with deviation log showing no architecture changes.
HYD-01	Confirm final pump curve overlay with rack/TCS pressure-drop curve and CDU internal dP for DI water and PG25.	CDU supplier; close with thermal-hydraulic model and factory curve.
HX-01	Confirm final HX manifold balance and approach-temperature margin at 2 MW / 500 GPM.	CDU supplier; close with HX flow balance calculation and FAT data.

ID	Open item	Owner / close condition
RELIEF-01	Confirm relief valve count/location/setpoint with AHJ and CDU OEM safety reviewer.	CDU supplier + ADC/AHJ; close with stamped safety basis.
MAT-01	Submit full wetted-material matrix; verify no nickel-plated brass on secondary side.	All suppliers; close with material cert package.
CTRL-01	Determine whether Redfish is native in controller or implemented by external bridge.	ADC + controls supplier; close with network architecture and cybersecurity review.
TCS-01	Map CDU-side TCS interface points to the separate ADC TCS header/branch package without changing TCS pipe sizing.	ADC + TCS supplier; close with interface control document.
SEIS-01	Replace generic Deschutes seismic basis with site-specific Louisiana parameters and stamped support design.	Structural PE; close with stamped analysis.
QA-01	Finalize FAT/validation sample quantities and test fluids.	ADC + CDU supplier; close with approved validation plan.

8. Source References

Source document	Sections used	How it was used
OCP Project Deschutes: Data Center Facilities v1.0.0, Effective Feb 26, 2026	Sections 3.1-3.3, 5.1-5.3, 6.1-6.7, 7.1-7.4, 8.1-8.2, 19-23.	Primary CDU design basis, pump/HX/valve/VFD first-source items, product requirements, validation, safety, serviceability, and controls.
ADC Solutions Modular TCS BoD/RFQ v1, May 19, 2026	Sections 1.1-1.3, 2.1-2.5, 4.6, 5.1-5.3.	CDU-to-rack boundary, secondary-side TCS interface assumptions, material policy, factory cleanliness, pressure ratings, TCS hydraulic context.
Deschutes Redfish, AI Control & Four-Tier Data Center Management v1.2, May 2026	Sections 2.2, 3.1-3.2, 4.4-4.5, 8.1-8.3.	Telemetry mapping, Redfish bridge concept, TCS rack-loop sensor mapping, PLC safety boundary, AI supervisory control context.
OCP Datacenter: Next-Generation ML Infrastructure Design Principles v0.5.0, Effective Feb 25, 2026	Sections 2, 3, and 3.2.	Data-center context, relationship of Deschutes CDU and Diablo sidecar to next-generation ML deployments.
OCP Diablo 400 Project: Rack and Power v0.5.2, Effective May 30, 2025	Sections 6-13.	Sidecar / high-density rack power context only; not used to alter the CDU internal power architecture.

Procurement note

This package is ready for RFQ discussion and supplier response. It should be released as a controlled preliminary BOM. A production BOM requires supplier CAD, P&ID, detailed assembly drawings, exact hose lengths, exact fastener counts, safety review, and validation signoff.